

Marco Mo — Data Engineer

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SUMMARY

Data Engineer with experience in designing and optimizing data pipelines for large-scale and complex datasets. Strong background in Python, SQL, and distributed processing (Spark), with hands-on experience in building reproducible workflows, data validation, and system performance optimization.

PROJECTS

Engineering Data Systems Database Project: quickSilver

- Optimized query execution and cardinality estimation for a C++ graph database.
- Improved performance of path queries using efficient traversal algorithms (BFS).
- Reduced memory usage and query latency, achieving top-3 performance in benchmarking.
- Focused on query optimization, data structures, and system-level performance.

Fraud Detection ML Pipeline (IEEE-CIS)

- Built an end-to-end data pipeline for fraud detection on 600k+ transactions.
- Designed data preprocessing and feature engineering workflows for high-dimensional data.
- Implemented validation strategies to ensure data integrity and prevent leakage.
- Simulated streaming data scenarios for real-time evaluation.

EXPERIENCE

Teaching Assistant — Data Challenges, Data Analytics, Calculus

Sep 2021 – Jul 2025

Eindhoven University of Technology (TU/e)

- Monitored group progress and guided teams using SCRUM-style meetings.
- Diagnosed underlying issues from students' answers and coached toward solutions.
- Led question-driven guidance to improve problem-solving and project quality.

Technical Lead – Student Team SimEnergy

Sep 2021 – Jul 2023

Eindhoven University of Technology (TU/e)

- Developed machine learning models for appliance-level energy disaggregation using time-series data.
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- Translated model outputs into dashboards for use in energy-focused events.

EDUCATION

MSc Data Science & Artificial Intelligence | Eindhoven University of Technology

2023 – 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration using Different Entropy Measures.

Honors Academy, AI Track | Eindhoven University of Technology

2021 – 2023

- Projects: Energy disaggregation, GAN image generation.

BSc Data Science | Eindhoven University of Technology & Tilburg University

2020 – 2023

- Thesis: Health AI Platform Text Classification.

SKILLS

Programming: Python, SQL, C++, R, Java

ML & Data: pandas, NumPy, scikit-learn, PyTorch, TensorFlow

LLM / GenAI: LangChain, Ollama, ChromaDB, Qdrant

Data Engineering / MLOps: Spark, MLflow, Docker, CI/CD (GitHub Actions), FastAPI

Visualization: Power BI, Plotly, Dash

Tools: Git

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)